Maxwell T. Elling

PhD Candidate, Atmospheric and Oceanic Sciences
University of Colorado Boulder
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Personal Site · Google Scholar · ResearchGate

Education

PhD in Atmospheric and Oceanic Sciences 2024–2027 (expected)
University of Colorado Boulder

MA in Climate and Society 2020–2021
Columbia University

BA in Financial Economics 2015–2019
Columbia University

Research Experience

Graduate Research Assistant 2024–present
University of Colorado Boulder

Scientific Programmer / Analyst 2023–2024
NASA Goddard Institute for Space Studies

Scientific Programmer 2021–2023
NASA Goddard Institute for Space Studies

Graduate Research Assistant 2021
International Research Institute for Climate and Society

Graduate Research Assistant 2021
Sabin Center for Climate Change Law

Graduate Research Assistant 2020–2021
Lamont Doherty Earth Observatory

Peer-Reviewed Publications

Published

9. **Elling, M.T.**, Karnauskas, K., Kowalcyk, M. et al.: Tropical oceans drive variability in soil moisture and malaria in Malawi. *Communications Medicine* (2026)

8. **Elling, M.T.**, Ruane, A.C., De Mel, M., Jonas, J., Kadam, S., Kiang, N.Y., Mezuman, K., Nazarenko, L., Pelaccio, N., Phillips, M., Romanou, A., and Rind, D.: An Impact-Driven Framework for Climate Model Evaluation. *Climatic Change* (2026). https://doi.org/10.1007/s10584-026-04157-w

7. Amooli, J.A., Miller, R.L., Tsigaridis, K., Chowdhury, S., Zhang, Y., Toolan, C.A., Ahmadi, S., Allen, R.J., **Elling, M.T.**, Ekman, A.M.L., Fraser-Leach, L., Griffiths, P.T., Keeble, J., Koshiro, T., Kushner, P.J., Lewinschal, A., Lund, M.T., MacRae, M., Makkonen, R., Merikanto, J., Nazarenko, L.S., Nabat, P., O'Donnell, D., Oshima, N., Persad, G., Rumbold, S.T., von Salzen, K., Samset, B.H., Swart, N.C., Takemura, T., Wilcox, L.J., and Westervelt, D.M.: Multi-Model Impacts of Dust on African Air Quality and Mortality under Regional and Global Anthropogenic Aerosol Changes. *Journal of Geophysical Research: Atmospheres* (2026). https://doi.org/10.1029/2025JD046135

6. Allen, R.J., Carson, T., Liu, W., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.**, Fraser-Leach, L., Griffiths, P., Keeble, J., Koshiro, T., Kushner, P., Lewinschal, A., MacRae, M., Makkonen, R., Merikanto, J., Nabat, P., Nazarenko, L.S., O'Donnell, D., Oshima, N., Paynter, D.J., Persad, G., Rumbold, S., Swart, N., Takemura, T., Tsigaridis, K., von Salzen, K., Westervelt, D.M., and Hassan, T.: AMOC weakening in response to global and regional reductions in aerosol emissions. *Environmental Research: Climate* (2026). https://doi.org/10.1088/2752-5295/ae63ef

5. Allen, R.J., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.**, Fraser-Leach, L., Griffiths, P., Keeble, J., Koshiro, T., Kushner, P., Lewinschal, A., MacRae, M., Makkonen, R., Merikanto, J., Nabat, P., Nazarenko, L.S., O'Donnell, D., Oshima, N., Paynter, D.J., Persad, G., Rumbold, S., Swart, N., Takemura, T., Tsigaridis, K., von Salzen, K., and Westervelt, D.M.: Decomposing the global and regional aerosol effective radiative forcing associated with strong versus weak air quality policies by Mid-21st Century. *Environmental Research: Climate* (2026). https://doi.org/10.1088/2752-5295/ae5418

4. Kramer, S.M., Karnauskas, K.B., **Elling, M.T.**, Zhang, L., Liu, H., Chen, Y., Amaya, D.J. et al.: ColdBlobMIP: A Multi-Model Assessment of the Atmospheric Response to the North Atlantic Warming Hole. *Geophysical Research Letters* (2025). https://doi.org/10.1029/2025GL117784

3. Nazarenko, L.S., Tausnev, N.L., and **Elling, M.T.**: Temperature and Precipitation Extremes Under SSP Emission Scenarios with GISS-E2.1 Model. *Atmosphere* (2025). https://doi.org/10.3390/atmos16080920

2. Yao, Y., Ducharne, A., Cook, B.I., De Hertog, S.J., Aas, K.S., Arboleda-Obando, P.F., Buzan, J., Colin, J., Costantini, M., Decharme, B., Lawrence, D.M., Lawrence, P., Leung, L.R., Lo, M.-H., Devaraju, N., Wieder, W.R., Wu, R.-J., Zhou, T., Jägermeyr, J., McDermid, S., Pokhrel, Y., **Elling, M.**, Hanasaki, N., Muñoz, P., Nazarenko, L.S., Otta, K., Satoh, Y., Yokohata, T., Jin, L., Wang, X., Mishra, V., Ghosh, S., and Thiery, W.: Impacts of irrigation expansion on moist-heat stress based on IRRMIP results. *Nature Communications*, (2025). https://doi.org/10.1038/s41467-025-56356-1

1. Samset, B.H., Wilcox, L.J., Allen, R.J., Stjern, C.W., Lund, M.T., Ahmadi, S., Ekman, A., **Elling, M.T.**, Fraser-Leach, L., Griffiths, P., Keeble, J., Koshiro, T., Kushner, P., Lewinschal, A., Makkonen, R., Merikanto, J., Nabat, P., Narazenko, L., O'Donnell, D., Oshima, N. et al.: East Asian aerosol cleanup has likely contributed to the recent acceleration in global warming. *Communications Earth & Environment* (2025). https://doi.org/10.1038/s43247-025-02527-3

In Preparation

Elling, M.T. and Karnauskas, K.B.: GCHI: A Global Climate Health Index Framework for Quantifying Climate Risks to Human Health. In prep.

Elling, M.T. and Karnauskas, K.B.: Evaluation of Future Global Climate Risks to Human Health. In prep.

Elling, M.T. and Karnauskas, K.B.: Evaluation of Future Climate Risks to Human Health in Africa. In prep.

Conference Papers

Cromp, S., Rabiei, B., **Elling, M.T.**, Herron, A., Hendrickson, M.: Climate PAL: Climate Analysis through Conversational AI

Technical Reports

Carr, E., McCabe, S., **Elling, M.T.**, and Martínez-Loustalot, P.: Emissions and Underwater Noise from LNG Carrier Vessels — A Study of the Potential Impacts of LNG Development on Marine Mammals in the Gulf of California.

Carr, E., McCabe, S., and **Elling, M.T.**: Well-to-Tank Carbon Intensity of Canadian LNG.

Carr, E., McCabe, S., and **Elling, M.T.**: The Impact of Ship Emission Fees on Mode Shift Potential in the United States.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: Well-to-Tank Carbon Intensity of European LNG Imports.

Carr, E., Winebrake, J.J., McCabe, S., **Elling, M.T.**, and Green, E.H.: Ocean-Going Vessel Decarbonization Technology Assessment.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: LNG as a Marine Fuel in the United States.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: LNG and Shipping in the Arctic.

Comer, B., McCabe, S., Carr, E., **Elling, M.T.**, Sturrup, E., Knudsen, B., Beecken, J., and Winebrake, J.J.: Real-world NOx emissions from ships and implications for future regulations.

Carr, E., McCabe, S., **Elling, M.T.**, and Winebrake, J.J.: Options for Reducing Methane Emissions from New and Existing LNG-Fueled Ships.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: Black Carbon Emissions from Ships in the Arctic 2019–2024.

Presentations (First Author & Presenter)

Elling, M.T., Karnauskas, K.B., Kowalcyk, M. et al., 2026: Tropical oceans drive variability in soil moisture and malaria in Malawi. European Geophysical Union, Annual Meeting 2026. *(Oral)*

Elling, M.T., Karnauskas, K.B., Kowalcyk, M. et al., 2025: Tropical oceans drive Malawi's malaria risk. American Geophysical Union, Annual Meeting 2025. *(Poster)*

Elling, M.T., Karnauskas, K.B., Kowalcyk, M. et al., 2025: Tropical oceans drive Malawi's malaria risk. Annual Earth System and Space Science Poster Conference 2025. *(Poster)*

Elling, M.T., Ruane, A.C., Rind, D. et al., 2024: Enhancing Earth System Model Development Through Climate Impact-Driven Model Evaluation. American Geophysical Union, Annual Meeting 2024. *(Poster)*

Elling, M.T., Ruane, A.C., Rind, D. et al., 2023: A Systematic Framework for Investigating Climate Impact-Drivers in NASA GISS ESM Configurations: Towards Improving ESM Development for Impact Applications. American Geophysical Union, Annual Meeting 2023. *(Poster)*

Peer Review

Climatic Change; *Environmental Research: Health*; *Geophysical Research Letters*

Teaching Experience

Teaching Assistant, Polar Ice and Climate — University of Colorado Boulder (Fall 2026)

Lectured on Snowball Earth and idealized modeling approaches for evaluating Snowball Earth conditions; created coursework and led laboratory sessions; co-developed experimental materials for measuring snow albedo and physical snow properties; guided students through hands-on experiments

Guest Lecturer, Environmental Policy — University of Colorado Boulder (Fall 2026)

Lectured on the intersections of climate science, policy, and society, and opportunities for strengthening connections across these domains

Mentoring

Zoë Pekarek, undergraduate researcher, University of Colorado Boulder (co-mentor; 2025)

Behrad Rabiei, intern, NASA GISS (co-manager; 2023)

Sonia Cromp, intern, NASA GISS (co-manager; 2023)

Shreya Chandra, intern, NASA GISS (manager; 2022)

Service

Co-founder, Climate Action Knowledge Exchange (2025–present)

Member, Atmospheric and Oceanic Sciences Curriculum Committee (2025/2026)

Member, Atmospheric and Oceanic Sciences Industry Night Committee (2025/2026)

Reviewer, Undergraduate Research Opportunities Program (UROP) proposals (2026)

Organizer, New Era Network for Societally-Integrated Climatology (2022–2023)

Participant, U.S.–Korea Young Climate Activists Exchange Program (2022)

Awards

Atmospheric and Oceanic Sciences Outstanding Service Award (2025/2026)

Annual Earth System and Space Science Poster Conference: 3rd Place (2025)

Technical Skills

UNIX/Linux and shell scripting

High-performance computing

Version control (Git)

PostgreSQL and Relational Database Management

Earth System Models: NASA GISS ModelE (code development, setting up and running experiments, CMORization/ESGF publishing/quality control); multi-model CMIP6 suite analysis (dozens of international models)

Open Source Software development: GCHI: A Global Climate Health Index

Programming languages: Python, Fortran, JavaScript, MATLAB, R

Professional Affiliations

American Geophysical Union (2022–present)